



Detailed Techno-Commercial Proposal

6.20 Kw Roof Top Solar System

Format 5 Verification Sample

Offer Verification Brand

Client: Format 5 Verification Client
Prepared For: Format 5 Verification Sample
Customer Contact: Verification Contact
Consumer Type: LT Consumer
Consumer No.: -
Prepared By: GitHub Copilot
Issued: 12 Apr 2026
Valid Till: 22 Apr 2026

From

Hi-Tech Solar

Bill To

Format 5 Verification Client
Contact: Verification Contact
Email: verification@example.com
Phone: 9876543210
Address: Verification Address, Dahod

Executive Summary

We are pleased to present our proposal for the installation of a 6.20 kWp solar power plant for Format 5 Verification Sample. This solution is designed to reduce electricity costs, lower carbon emissions, and provide long-term energy reliability through a complete EPC scope covering design, procurement, installation, commissioning, and post-installation support.

Description of Services

- Engineering:
 - Detailed site survey, feasibility analysis, and complete electrical and structural design.
- Procurement:
 - Solar PV modules, inverters, structures, cables, connectors, switchgear, and protection systems compliant with BIS and IEC standards.
- Construction & Installation:
 - Civil works, structure installation, module mounting, cabling, earthing, lightning protection, and grid synchronization.
- Testing & Commissioning:
 - Pre-commissioning checks, energization, performance validation, and authority coordination.
- Training & Support:
 - Basic operator training, handover support, and post-installation technical assistance.
- Documentation & Approvals:
 - Assistance for approvals, as-built records, warranty papers, and handover documentation.

Technical Proposal

Parameter	Description	Remarks
System Size	6.20 Kw	-
System Type	On Grid	Verification Solar Inverter
Required Area	310 Sqft Minimum	50 sqft per kW



Technical Proposal

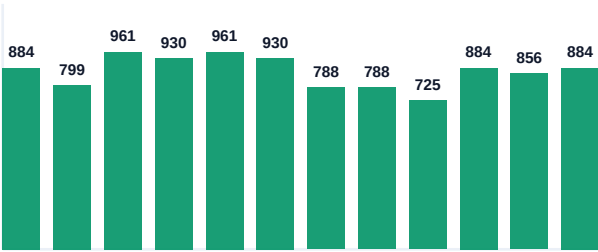
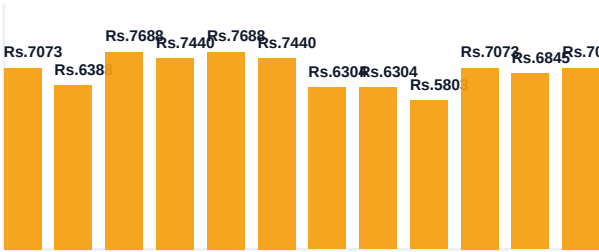
Parameter	Description	Remarks
Module Type	620Wp Verification Solar Module	-
Module Make & Rating	DemoSolar	30 years linear power warranty
Inverter Type	Verification Solar Inverter	On-grid inverter for PDF verification
Inverter Make & Rating	DemoInvert	-
Inverter Warranty Coverage	5 years from manufacturer	-
Type Of Module Mounting Structure	Verification Mounting Structure	Wind Speed Resistance as per IS 875: up to 100 kmph
Structure Height	15 ft at South, 17 ft at North	-
Array Layout	South Facing, 14 Tilt, Portrait orientation	-
Type of ACDB Box	-	-
Type of DCDB Box	-	-
Type of Lightning Arrestor	-	-
Type of Earthing Road	-	-
Monitoring System	Remote Monitoring System (Wi-Fi/GPRS), Web & Mobile Interface	-
Net Metering Provision	System compatible and provision for Net Metering as per DISCOM norms	-
Expected Generation (Unit/Day)	28	4.59 units/day/kW
Approvals & Compliance	MNRE, DISCOM, CEIG, IEC/BIS Standards	-
Project Completion Timeline	8 - 10 Weeks from advance and site readiness	-



System Generation & Revenue Details

Solar Roof Top Size 6.20 Kw Calculated system size	Generation / Day 31 Unit System size x rounded avg. generation	Yearly Saving Rs.83,119.68 Sum of monthly savings
1st Year Generation 10390 kWh Sum of monthly generation	25 Year Saving Rs.20,77,992.00 Yearly saving x 25	

Predicted Monthly Generation & Savings

Predicted Monthly Generation (kWh) <table><thead><tr><th>Month</th><th>Generation (kWh)</th></tr></thead><tbody><tr><td>JAN</td><td>884</td></tr><tr><td>FEB</td><td>799</td></tr><tr><td>MAR</td><td>961</td></tr><tr><td>APR</td><td>930</td></tr><tr><td>MAY</td><td>961</td></tr><tr><td>JUN</td><td>930</td></tr><tr><td>JUL</td><td>788</td></tr><tr><td>AUG</td><td>788</td></tr><tr><td>SEP</td><td>725</td></tr><tr><td>OCT</td><td>884</td></tr><tr><td>NOV</td><td>856</td></tr><tr><td>DEC</td><td>884</td></tr></tbody></table>	Month	Generation (kWh)	JAN	884	FEB	799	MAR	961	APR	930	MAY	961	JUN	930	JUL	788	AUG	788	SEP	725	OCT	884	NOV	856	DEC	884	Predicted Monthly Savings (INR) <table><thead><tr><th>Month</th><th>Savings (INR)</th></tr></thead><tbody><tr><td>JAN</td><td>Rs.7073</td></tr><tr><td>FEB</td><td>Rs.638</td></tr><tr><td>MAR</td><td>Rs.7688</td></tr><tr><td>APR</td><td>Rs.7440</td></tr><tr><td>MAY</td><td>Rs.7688</td></tr><tr><td>JUN</td><td>Rs.7440</td></tr><tr><td>JUL</td><td>Rs.630</td></tr><tr><td>AUG</td><td>Rs.6304</td></tr><tr><td>SEP</td><td>Rs.580</td></tr><tr><td>OCT</td><td>Rs.7073</td></tr><tr><td>NOV</td><td>Rs.6845</td></tr><tr><td>DEC</td><td>Rs.7073</td></tr></tbody></table>	Month	Savings (INR)	JAN	Rs.7073	FEB	Rs.638	MAR	Rs.7688	APR	Rs.7440	MAY	Rs.7688	JUN	Rs.7440	JUL	Rs.630	AUG	Rs.6304	SEP	Rs.580	OCT	Rs.7073	NOV	Rs.6845	DEC	Rs.7073
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Avg. Unit / Day 4.59	Total Days 365	1st Year Generation 10390 kWh	Yearly Saving Rs.83,119.68
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Note Predicted generation is indicative only. It may vary depending on weather conditions and module condition. Savings are calculated at the Year 1 tariff.

A Billing of Quantities

Item Head	Description	Qty	Rate	Amount
SOLAR MODULE	- Mono PERC module for PDF verification	10	Rs.11,200.00	Rs.1,12,000.00
SOLAR INVERTER	- On-grid inverter for PDF verification	1	Rs.42,000.00	Rs.42,000.00
SOLAR STRUCTURE	- Hot dip galvanized mounting structure	1	Rs.1,28,000.00	Rs.1,28,000.00



Scope of Work Matrix

Sr	Work Item / Activity	Responsibility	Remarks
1 Project Development & Engineering			
1.1	Feasibility Study (technical & financial)	Consumer with EPC Contractor	Can be jointly discussed with EPC.
1.2	Site Survey & Assessment	EPC Contractor	Includes structural and electrical feasibility.
1.3	System Design & Engineering Drawings	EPC Contractor	Electrical SLD, layout, shadow analysis, etc.
1.4	Statutory and Safety Compliance (design level)	EPC Contractor	Must follow MNRE, DISCOM, CEIG norms.
1.5	Land Levelling and Grading/Tree Cutting	Consumer	In Consumer Scope
1.6	Cable Trenches	Consumer	Existing Trenches considered
1.7	Control Rooms (RCC/PEB)	Consumer	Existing Covered Space in Control Room considered
1.8	Water Drainage	Consumer	In Customer Scope. Existing Drainage Considered
1.9	Water and Electricity	Consumer	Supply of Free Water and Free Electricity in Customer Scope. during I&C and O&M (If Appl.)
2 Procurement & Supply			
2.1	Procurement of Solar Modules (approved make)	EPC Contractor	Must meet MNRE ALMM list and BIS norms.
2.2	Procurement of Inverters	EPC Contractor	Must meet DISCOM & MNRE specs.
2.3	Procurement of Mounting Structures	EPC Contractor	Corrosion-resistant (hot dip galvanized).
2.4	Cables, Junction Boxes, Earthing Material	EPC Contractor	IS-compliant cables and materials.
2.5	Transportation to Site	EPC Contractor	Delivered to consumer site.
3 Installation & Construction			
3.1	Module Mounting Structure Installation	EPC Contractor	Based on roof type (RCC, sheet metal, etc.).
3.2	Module Installation	EPC Contractor	Including alignment and clamping.
3.3	Inverter & Electrical Panel Installation	EPC Contractor	Includes ACDB, DCDB, SPD, meters, etc.
3.4	Earthing, Lightning Protection System	EPC Contractor	Minimum 2 earth pits per inverter.
3.5	Cable Laying and Termination	EPC Contractor	As per standard cable routing practices.
3.6	ACDB to Client LT Panel (AC Cabling)	EPC Contractor	All AC Cables above 16 mm ² : 1.1KV Al conductor (Class-2), GS Armoured, XLPE Insulation, PVC outer sheath Cable.
3.7	Client LT Panel to Client HT Panel (AC Cabling)	DISCOM	N/A



Scope of Work Matrix

Sr	Work Item / Activity	Responsibility	Remarks
3.8	Cable Tray/ Conduits with supporting structures	EPC Contractor	Conduits have been considered for DC/AC cable Routing
3.9	HT Panel	N/A	Evacuation considered at LT Panel of same building @415V
3.10	Step-Up Transf. & Accessories	N/A	Evacuation considered at LT Panel of same building @415V
3.11	Module Cleaning System	Consumer	In Consumer Scope
3.12	Hand Rail	Consumer	Handrail/Parapet Walls on RCC Roofs in Customer Scope.
3.13	Staircase/Monkey Ladder	Consumer	Site Access (Staircase) in Customer Scope
3.14	PV-DG Controller	Consumer	Not Considered in any form ie . RPR/ Fuel Saver PLC Based PVDG Controller, etc.
3.6	Civil Works (if required)	EPC Contractor	Pedestals, cable trenching (if needed).
4 Grid Connection & Net Metering			
4.1	Liaisoning with DISCOM for Grid Approval	EPC Contractor (mostly)	Consumer may assist in document signatures.
4.2	Application for Net Metering	EPC Contractor (with Consumer)	Consumer's documents like PAN, Aadhaar needed.
4.3	Procurement of Bi-directional meter, Modem And Meter Box	Consumer (With EPC Contractor)	Only Applicable In HT Consumers.
4.4	Installation of Net Meter	DISCOM (coordinated by EPC)	EPC will ensure DISCOM visit and coordination.
4.5	CEIG Approval (if required)	EPC Contractor (with Consumer)	Required for capacity >10kW in Gujarat.
5 Testing & Commissioning			
5.1	System Testing (insulation, polarity, continuity, etc.)	EPC Contractor	Complete testing prior to energization.
5.2	Synchronization with Grid	EPC Contractor	In coordination with DISCOM.
5.3	Final Commissioning	EPC Contractor	Post-approval and net meter installation.
6 Documentation & Handover			
6.1	Submission of Final Drawings & Technical Documents	EPC Contractor	Includes SLDs, datasheets, manuals.
6.2	Warranty Certificates	EPC Contractor	For modules, inverters, BOS items.
6.3	O&M Manual & Training to Consumer Staff	EPC Contractor	Brief on periodic maintenance & troubleshooting.
7 Operation & Maintenance			



Scope of Work Matrix

Sr	Work Item / Activity	Responsibility	Remarks
7.1	O&M Responsibility (5 year free O&M typically)	EPC Contractor	Optional extension through AMC.
7.2	Performance Monitoring System	EPC Contractor	Remote monitoring (Wi-Fi/GPRS) for real-time data.
8 Commercial Aspects			
8.1	Project Financing	Consumer	Unless provided by EPC under RESCO/lease.
8.2	Insurance of Plant (during & post-installation)	Consumer	in consumer scope
8.3	Payment Terms	Consumer	As per contract milestones.
8.4	Transportation	EPC Contractor	Include in EPC Contractor Scope.
8.5	Payment Terms	Consumer	As per contract milestones.

Solar Roof Top System Pricing	
Solar Power Generating System With GST@0.00%	Rs.2,82,000.00
Solar System & Installation Cost	Rs.2,82,000.00
Tax / GST	Rs.0.00
Total Amount / Total Kw	Rs.45,483.87
Registration Charges	Rs.0.00
Net Metering Charges	Rs.0.00
Meter / Modem Charges	Rs.0.00
Grand Total	Rs.2,82,000.00

Solar Plant Installation : Procedure & Time frame

Steps	Procedure	Description	Time Period (Approx)
Step-1	GEDA Registration	Start registration process by collecting required documents.	1-2 working days
Step-2	Document Verification	Verification of uploaded documents after registration.	4-5 working days
Step-3	Feasibility Approval	DISCOM gives feasibility approval and issues the solar meter estimate after document verification.	10-15 working days
Step-4	IFP Drawing Approval	Applicable above 10 kW. Electrical drawing is prepared and uploaded on the IFP portal for execution approval.	15-20 working days
Step-5	Execution Work	Plant execution starts after all required approvals are in place.	15-20 working days
Step-6	CEIG Inspection Approval	Applicable above 10 kW. After installation, CEIG inspection is applied for and the inspector verifies the site physically.	15-20 working days
Step-7	Net Metering Process	After CEIG approval, the net metering file is prepared and submitted to DISCOM for meter installation.	30-45 working days
Step-8	Commissioning & Testing	Testing is carried out after installation and CEIG approval.	1-2 working days



Solar Plant Installation : Procedure & Time frame

Steps	Procedure	Description	Time Period (Approx)
Step-9	Completion of Project	After testing, the plant is handed over to the client.	1-2 working days
Note	The procedure and timeframe mentioned above are approximate and may vary depending on weather conditions, document processing delays, payment delays, and approvals from the concerned authorities.		

Solar Plant ROI Calculator

Parameter	Unit	Value	Notes
Project Capacity	kW	6.20	Size of plant
Installation Cost	INR	Rs.2,82,000.00	Total EPC cost
Average Daily Generation	kWh / kWp / day	4.00	Based on site data
Average Yearly Shutdown	Days	30	Plant or grid-side maintenance downtime
Electricity Tariff (Year 1)	INR / kWh	Rs.8.00	Current grid rate
Tariff Escalation	%	3.00	Expected yearly increase
Annual Power Degradation (After 1st Year)	%	2.00	Module efficiency drop after Year 1
Annual Power Degradation (From 3rd Year onward)	%	0.50	Module efficiency drop from Year 3 onward
O&M Cost (Year 1)	% / INR	1.00 Rs.2,82,000.00	Year 1 maintenance cost
O&M Cost Escalation	%	3.00	Inflation in O&M
Project Life	Years	30	Standard plant life used for ROI projection
Year 1 Generation 8308 kWh	Year 1 Gross Savings Rs.66,464.00		Year 1 Net Savings -Rs.2,15,536.00
Estimated Payback Beyond projection	Lifetime Net Savings -Rs.1,32,11,475.89		

Payment Stages

Stage	Milestone / Remarks	%	Amount
Stage 1: Booking Amount	At the time of order confirmation. Ensures order confirmation and documentation.	10%	Rs.28,200.00
Stage 2: Approval Process	After registration and feasibility approval. Covers registration and initial processing work.	30%	Rs.84,600.00
Stage 3: Installation	After delivery of materials and start of installation work. For procurement of materials and labor costs.	30%	Rs.84,600.00
Stage 4: Testing & Inspection	After successful installation and inspection approval. Covers inspection, testing, and net metering processes.	20%	Rs.56,400.00
Stage 5: Project Handover	On successful commissioning and handover of the solar plant. Final settlement after completion of the project.	10%	Rs.28,200.00



Terms and Conditions

- Price validity:
- 15 days from the quotation issue date unless revised in writing.
- Payment terms:
- 70% against material dispatch and 30% against successful installation and handover.
- Execution timelines depend on site readiness, approvals, and material availability.
- Any civil, electrical, or statutory scope outside the listed BOQ will be billed separately after approval.
- Taxes are included as shown above and final billing will follow the applicable GST at invoicing.

Bank Details	Required Documents
Bank Name: AU Small Finance Bank A/C Name: Hi-Tech Solar Solution A/C Number: 1107865152531445 A/C Type: Current Account IFSC Code: AUBL0002158 Branch: Dahod	- Latest Electricity Bill - Ownership proof - Aadhar & PAN card of authorized person - PAN card & cancelled cheque of firm - Passport size photo of authorized person - GST certificate - Undertaking on stamp paper - Authorization letter

Authorization

Client Acceptance Signature / Date	For Hi-Tech Solar Authorized Signatory
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